

MISTRAL AI · VIBE HACKATHON · PARIS · JULY 18, 2026



Vibe Pulse

Give your agent eyes · live ML training & data
visualization, right in the terminal.

THE GAP

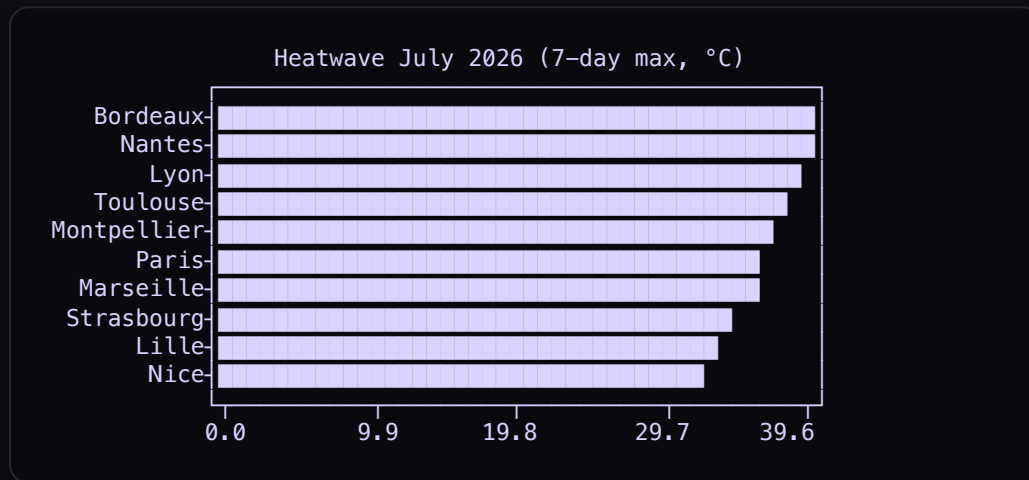
Terminal-first ML devs still **leave the terminal** to see a curve

- ▶ Vibe **Work** charts your data in Canvas. Vibe **Code** (CLI): nothing.
- ▶ Jupyter or TensorBoard just to check convergence = context-switch tax.
- ▶ No terminal coding agent does this natively.
None.



LIVE DEMO · REAL DATA

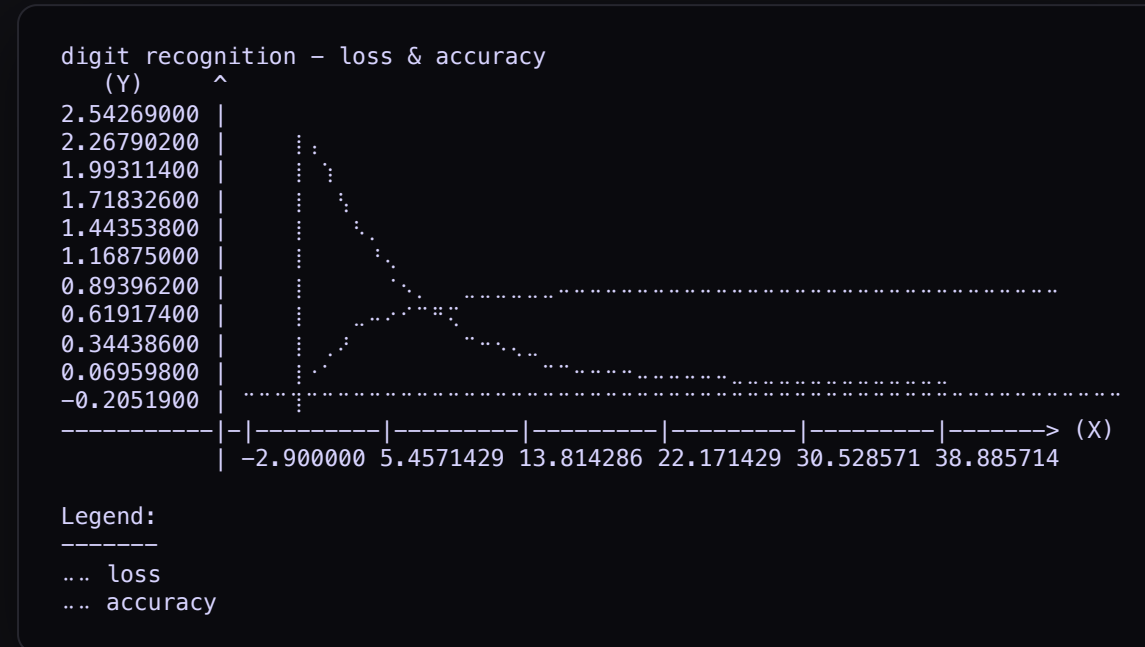
Any data, straight in the terminal



- ▶ CSV, JSON streams, **Excel**, **SQLite** · one command each.
- ▶ The July 2026 heatwave, charted from a SQL query.
- ▶ Same charts, in color, **inside the Vibe conversation**.

IT ACTUALLY WORKS

Machine learning, **fully** in the terminal

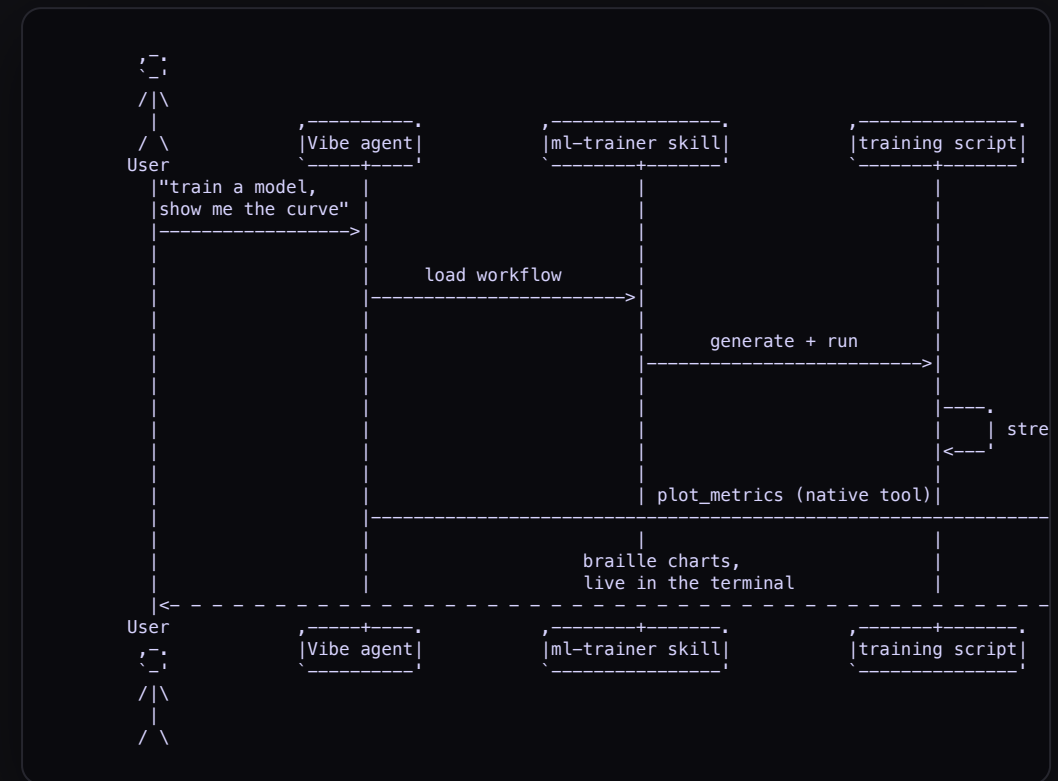


Digit recognition to **98 %**, curves drawn live epoch by epoch · confusion heatmap and the misclassified digits, as images, in the conversation.

UNDER THE HOOD

Every Vibe extension surface, **used**

- ▶ **Skill** · ml-trainer orchestrates the loop
- ▶ **Native tool** · plot_metrics, 8 chart kinds, colored TUI widget
- ▶ **Provider** · Vibe inference on Qualcomm Cloud AI 100
- ▶ **Hooks** · post_tool telemetry + pre_tool secrets-guard



WHO IT SERVES

Three engineers, **one terminal**



Data scientists / ML engineers

live training curves, confusion heatmaps, run tracking & comparison, agent-driven sweeps · no Jupyter, no TensorBoard.



Software engineers

PlantUML architecture diagrams as ASCII art in the console (Mermaid next), agent telemetry · dev workflow, in place.



Embedded systems engineers

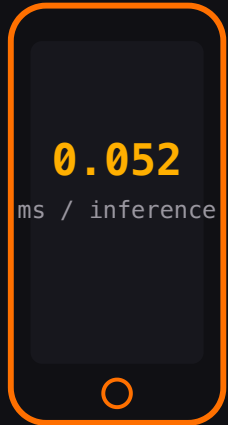
compile + profile on real Snapdragon from the same session — 0.052 ms measured, not estimated.

W&B needs a server and a browser. **We don't.**

- ▶ **Run tracking** · every run logged to `runs/index.jsonl`
- ▶ **Run comparison** · `vibe-plot --compare` overlays any metric across experiments
- ▶ **Agent-driven sweeps** · “try 3 learning rates and compare” · no sweep server, the agent is the orchestrator
- ▶ **ASCII PlantUML diagrams** · architecture & dependency graphs without leaving the terminal
- ▶ **Canvas-to-CLI** · a Vibe Work skill converted unchanged (doc, table, chart twins)
- ▶ **vibe-arena** · LMArena-style model comparison in the console: Mistral vs OpenAI, latency & tok/s charted
- ▶ **Any data** · CSV, JSON streams, Excel, SQLite (`--sql`) · and Canvas-style tables (`--table`)

REAL SILICON

Trained in the terminal, **validated on Snapdragon**



0.052 ms

on-device inference
Samsung Galaxy S24+ (AI Hub)

24 MB

peak memory on device

Cloud AI 100

Vibe's own inference runs on
Qualcomm silicon (Cirrascle)

PR-ready upstream

- ▶ Draft PR **mistralai/mistral-vibe #920**: a native `plot_metrics` builtin
- ▶ Zero new dependencies · built on `render_braille` ...
- ▶ ...the primitive that animates **Vibe's mascot**. First product feature for the cat. 🐱
- ▶ Found & documented a real API pitfall (braille dot-index overflow) along the way

ROADMAP

The terminal as a **first-class** data science surface

- ▶ **Richer TUI** · textual-plotext RGB widgets, rich-pixels sample images, zoom/pan
- ▶ **Complex models** · CNNs, real datasets, fine-tuning, GPU runs · same live workflow
- ▶ **Fully local** · Ollama/llama.cpp provider is one config line away; a `/local` switch is in the works
- ▶ **Canvas-to-CLI** · full coverage (first Work skill converted today)
- ▶ **Deeper analysis** · dataset profiling, correlations, drift monitoring

VIBE PULSE

Give your agent **eyes.**

Works today · a skill, a native tool, a provider, a hook.

github.com/wilfred-dore/vibe-pulse · PR [mistralai/mistral-vibe](#)
#920